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Shotgun metagenomic sequencing on skin microbiome indicates dysbiosis exists prior to the onset of atopic dermatitis.

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Abstract

Background: While the microbiome is increasingly seen as a targetable contributor to atopic dermatitis (AD), questions remain as to whether the dysbiosis is secondary to diseased skin or if it predates symptom onset. Previous work has evaluated how the skin microbiome changes with age and established the influence of factors like delivery mode and breastfeeding on global microbiome diversity. However, these studies were unable to identify taxa which predict subsequent AD.

Methods: Skin swab samples were collected from the first week of life for 72 children in the neonatal intensive care unit (NICU) at a single site hospital. Participants were followed for three years to determine their health status. We applied shotgun metagenomic sequencing to assess the microbiome differences between 31 children who went on to develop AD and 41 controls.

Results: We identified that subsequent development of AD was associated with differential abundance of several bacterial and fungal taxa as well as several metabolic pathways, each of which have been previously associated with active AD.

Conclusions: Our work provides evidence of reproducibility for the previously reported dysbiotic signatures predating AD onset while also expanding prior findings through the first

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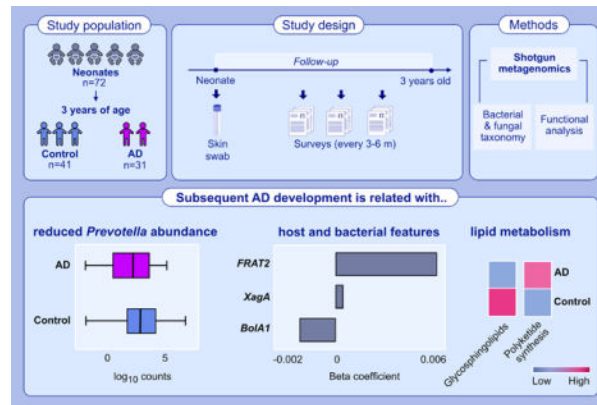
Author's Contributions

PPC, JZ and SL performed data analysis and prepared figures. SD provided fungal analysis. VD, RB, RBak, SB, and AS assisted in sample collection and data analysis. IAM and SKH wrote the manuscript text. All authors reviewed the manuscript.

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use of metagenomic assessment prior to AD onset. While extrapolation of our findings beyond the pre-term, NICU cohort is limited, our findings add to the evidence that the dysbiosis associated with AD pre-dates disease onset rather than reflect a secondary consequence of skin inflammation.

Graphical Abstract



- Using a larger cohort, this study reproduces the prior findings indicating that the dysbiosis associated with AD predates diagnosis.
- Expansion to use of shotgun metagenomics identified *Prevotella*, *Candida*, and lipid metabolism of *Proteobacteria* as additional biomarkers of subsequent onset of AD.
- This work may foster primary prevention by using the microbiome signature from the first week of life to identify high risk newborn.

Abbreviations: AD, atopic dermatitis; *BolA1*, BolA Family Member 1; *FRAT2*, frequently rearranged in advanced T-cell lymphomas 2; *XagA*, *Proteobacteria* quorum sensing and biofilm regulator

Keywords

Microbiome; atopic dermatitis; eczema; dysbiosis; *Staphylococcus*; *Malassezia*

Background:

The microbiome is increasingly seen as a targetable contributor to atopic dermatitis (AD)[1]. However, questions remain as to whether AD-associated dysbiosis is secondary to diseased skin or if it predates symptom onset in ways that may offer opportunities for primary prevention[2]. To address this, one study used longitudinal 16S rRNA sequence taxonomy in 10 children with AD and 10 matched controls[2]. The researchers sampled four body sites on day 2, month 2, and month 6 to contrast with AD diagnosis at 1 year of age. The work successfully described site and age-specific differences. The one predictive signal detected was a reduction in the relative abundance of *Staphylococcus* on the antecubital fossae at 2 months, which was associated with the four patients that showed persistent disease at 1 year but did not predict general AD risk[2]. More recent work collected swabs from 346 newborns on day 1 of life and was able to follow 281 infants for 12 months to assess change in 16S rRNA profiles[3]. The authors identified taxa such as *Streptococcus*

and *Bacilli* as predictors of dry skin, however no bacterial taxa associated with a formal diagnosis of AD[3]. Thus, we aimed to expand on this prior finding by deploying shotgun metagenomic sequencing on a cohort of 72 neonates admitted to the neonatal intensive care unit at the Inova Fairfax Hospital (Fairfax, VA) and followed until approximately 3 years corrected age (see supplemental methods for details of cohort). We compared skin swabs taken from behind the ear during the first week of life to contrast the microbiome of the 31 neonates who later reported developing AD (at least half of which were confirmed by providers) versus the 41 whom did not report development of AD. Although the microbiome signatures were similar overall, we identified reduced abundance of *Prevotella* was the taxonomic feature that was most predictive of subsequent development of AD. Fungal analysis indicated a role for *Candida* species while functional analysis indicated lipid metabolism from *Proteobacteria* were important for assessing AD risk. Taken together these results provide needed reproduction of prior research and lay the groundwork for future work that may be able to use microbiome signatures to identify high risk newborns.

Methods:

Patient cohort

Neonates who spent time in the Inova Fairfax Neonatal Intensive Care Unit (NICU) were enrolled in an observational longitudinal microbiome cohort study. The study was Institutional Review Board approved (WCG IRB 1300205) and parental informed consent was obtained. Neonates were enrolled within the first week of life and had an anticipated stay in the NICU of >5 days. While in the NICU detailed demographic and clinical data was collected along with samples for microbiome analysis (including skin swabs and stool). Specifically for the skin, a sterile cotton swab was moistened with sterile saline and rolled behind the neonate's ear in a standardized manner.

After discharge from the NICU, follow-up surveys reporting health, illnesses and diet were collected approximately every 3–6 months until approximately 3 years of age (corrected for gestational age). For this current study, neonates from the larger cohort were included if they had a skin swab available from the first week of life and if the family completed the longitudinal surveys.

For the 31 infants in this study who had AD reported on their longitudinal surveys, an additional questionnaire was sent to the family to obtain more information about the AD diagnosis; this included questions from the validated Patient Oriented Eczema Measure, POEM (<https://www.nottingham.ac.uk/research/groups/cebd/resources/poem.aspx>).

Microbiome metagenomic sequencing and analysis

Sample collection: Skin swabs were collected using cotton swabs in sterile saline, taken from the behind the neonates ear, and then underwent DNA extraction (CosmosID, Germantown MD).

DNA Extraction, Library Preparation and Sequencing: DNA from skin swabs was isolated using the ZymoBIOMICS MicroPrep kit, according to the manufacturer's protocol. Extracted DNA samples were quantified using the GloMax Plate Reader System (Promega)

using the QuantiFluor® dsDNA System (Promega) chemistry. DNA libraries were prepared using the Nextera XT DNA Library Preparation Kit (Illumina) and IDT Unique Dual Indexes with total DNA input of 1ng. Genomic DNA was fragmented using a proportional amount of Illumina Nextera XT fragmentation enzyme. Unique dual indexes were added to each sample followed by 15 cycles of PCR to construct libraries. DNA libraries were purified using AMPure magnetic Beads (Beckman Coulter) and eluted in QIAGEN EB buffer. DNA libraries were quantified using Qubit 4 fluorometer and Qubit™ dsDNA HS Assay Kit. In order to do normalization sample libraries were pooled to target a depth of 20M reads per sample based on their Qubit-measured library concentrations, however due to differences in library quantity, quality, and sequencing variation, there was a range of depths for sequencing. Libraries were then sequenced on an Illumina NovaSeq 6000 platform 2×150bp.

Quality Assessment: Upon data arrival, raw data are backed up to Amazon AWS and run through fastqc and a multiqc report is generated. The multiqc report is checked to ensure read depth thresholds are met, and that there are no abnormalities with read quality. Regarding quality scores, CosmosID reviews the sequence quality histogram for any samples marked as “failed” (indicated by red lines) within the fastQC report, and do not deliver those samples if they are not previously marked as “at risk” due to low library concentration (<0.75 ng/uL via qubit). All extraction batches have an associated positive and negative process control that are processed with the samples from extraction to sequencing and there was no evidence of contamination in these samples.

Metagenomic data processing and statistical analysis: Raw reads were trimmed for quality by using Sickle (Vesion-1.33). Bowtie2 (version –2.4.2) was then used to run mapping against the human genome and the subsequent aligned reads were removed to get rid of host reads. The phylogenetic profile of each subject was determined using Kraken (version –1.2.0). For carrying out functional analysis gene calling was performed with Prodigal (version –2.6.3) and called genes were aligned against NCBI-nr database using DIAMOND (version –2.0.11.149). Finally, the functional analysis was performed by using MEGAN (version –6.13.3). Both the taxonomic and functional count tables were then loaded to Microbiomeanalyst and after rarefying the data to the minimum library size and data scaling different statistical analysis were performed and plots were generated. P values for alpha diversity and non-metric multi-dimensional scaling (NMDS) was calculated by using Mann Whitney/Kruska wallis option in Microbiomeanalyst while NMDS was generated by using bray-curtis dissimilarity matrix and significance between the groups was calculated by using ANOSIM test. Heatmap plot for functions (Fig. 2) was generated in R by using pheatmap package. Differential expression at the different the taxonomic and functional levels were calculated by using edge R by keeping value of $P < 0.05$ as threshold of significance. Elastic net logistic regression was performed using the glmnetUtils package (Ooi H, 2021) using R package version 4.2.1. The data sets generated and/or analyzed in the current study are available in the NCBI SRA repository under submission ID SUB11643006 and BioProject ID PRJNA854811

Results:

Birth mode and demographics did not associate with subsequent AD development

There were no differences in baseline demographic, day of collection, or clinical features between the neonates that went on to develop AD and those that did not (Table 1). Furthermore, consistent with the recent meta-analysis[4], birth by Caesarian section was not a risk factor for subsequent AD (Table 1). The mean age of AD diagnosis was 15 months (range 4 – 42 months). The subset of patients (15 of 31) that provided post-hoc information on disease distribution and severity suggested the cohort was typical for pediatric AD with the expected body distribution on the arms, legs, and face that was primarily managed with emollients and topical steroids (Supplementary Table 1).

Reduced *Prevotella* abundance predicted subsequent AD development

Similar to the prior reports[2], taxonomic analysis of swabs taken prior to diagnosis revealed overall similarities between children that were eventually diagnosed with AD and those that did not develop AD (Fig. 1A–B). There were no significant differences in either alpha or beta diversity, even when separating those with provider confirmed diagnoses from the self-reported patients (Fig. 1C–D). Reduction in relative abundance of *Staphylococcus* in those who went on to manifest AD did not meet statistical significance (Fig. 1E) and random forest assessment of the taxa instead identified *Actinobaculum* as the genus most distinguishing physician confirmed AD from controls (Fig. 1F); the relative abundance of *Actinobacteria* has been previously identified as lower in established AD patients than controls in at least one prior study[5]. Further groupwise taxonomy revealed that those with an eventual diagnosis of AD demonstrated significant reductions in the relative abundance of several taxa previously linked to active AD[6–8] such as *Prevotella*, a significant overabundance of *Fusobacteria*, but no significant difference in *S. aureus* abundance (Fig. 1G). In fact, no significant differences were seen on species or in isolate level analysis (not shown).

AD has also been associated with alterations in the fungal commensal taxa, with particular associations with differences in *Malassezia* species and *Candida*[9]. Here we found significantly increased levels of *Candida* and *Saccharomyces* in those that went on to develop AD when assessing the AD cohort in general (Fig. 2A–B; $p < 0.001$ for each genre), however significance was lost when considered provider versus self-diagnosis (Fig. 2C). Given that our samples were collected from the ear we performed a post-hoc analysis for the 15 AD-subjects who provided disease location data (Supplementary Table 1). We found that the 7 subjects that reported AD with face involvement had similar dysbiotic taxonomy as the overall cohort (Fig. 2D) with a site-specific enrichment in reduced abundance of *Prevotella* (Fig. 2E).

Functional analysis indicates difference in lipid metabolism predict subsequent AD development

We next analyzed the sequence data for metabolic function differences by Kyoto Encyclopedia of Genes and Genomes (KEGG) metabolic pathways. Like taxonomy, level 3 KEGG analysis revealed overall similarity in metabolic pathways (Fig. 3A–B), alpha

diversity (Fig. 3C) and beta diversity (Fig. 3D). The largest, and only false discovery rate (FDR)-adjusted significant difference was in polyketide biosynthesis (log2 fold change (FC) 1.3 for AD versus control); differences in the glycosphingolipid biosynthesis ganglio series were the second largest (log2FC 0.69) however the significance for this pathway was lost under FDR adjustments (Fig. 3E–F). Specific genetic processes significantly differed for those with an eventual diagnosis of AD (Fig. 3G; Fig. S1). The largest differences were seen in two enzymes in the LPS biosynthesis pathway (Fig. 3G; Fig. S1): UDP-3O3 hydroxymyristoyl N acetylglucosamine deacetylase (log2FC 4.48, FDR 1.6e-8) and ethanolaminephosphotransferase (log2FC -4.51, FDR 2.7e-5). This is consistent with the reported importance of lipid metabolism in AD[10].

Logistic regression identified host and bacterial features predicting AD development

Finally, we combined the bacterial taxa, fungal taxa, alpha diversity, gender, gestational age, delivery mode, and metabolic expression into a single elastic net logistic regression. The top distinguishing features under the combined analysis included overexpression of the *Wnt-β-catenin* related gene, frequently rearranged in advanced T-cell lymphoma 2 (*FRAT2*), overexpression of the *Proteobacteria* quorum sensing and biofilm regulator of XagA, along with reduced expression of BolA-like protein 1, which mitigates glutathione deficiency in *Proteobacteria*[11] (Fig 4A–B). Repeating the analysis looking for factors which distinguish physician confirmed AD further suggested that XagA showed the strongest association with subsequent AD (Fig. 4C–D).

Conclusions:

Our results identified an increased number of taxa beyond *Staphylococcus* which may be risk factors for AD development[2], while also demonstrating the reproducibility in prior literature that is often undervalued in microbiome research[12]. The addition of functional analysis suggested that lipid defects, inflammation, and oxidative stress signals previously associated with host pathways of AD [10, 13, 14] may predate symptom onset and possess corollaries in the metabolism of the microbiome. Assessing which environmental exposures disrupt our identified taxa and microbial metabolic pathways may further elucidate the origins of the dysbiosis associated with AD.

These enhanced results may reflect this increased power and metagenomic approach of our study, our larger sample size, or the longer follow up of 3 years rather than 12 months[3]. However, our results may also reflect limitations in our cohort which included babies born prematurely and exposed to the unique environments of the NICU. Although not associated with AD diagnosis, the rate of cesarean delivery was higher in our cohort than reported US averages [15] which resulted in higher rates of peripartum antibiotic exposure [16]. Although some studies have suggested a lower risk of AD in premature infants compared to full term infants [17], given our entire cohort shared the same prematurity and delivery mode “exposures” in this study, the factors associated with the development of AD risk within this cohort remain of value. Our study was also limited to a single study site and single body sample site and thus may not be fully indicative of the geographic distribution of AD symptoms.

While this work is the first metagenomic assessment of perinatal dysbiosis predating AD, to derive a predictive model from this data future work must prospectively correlate results with disease severity, body site involvement, and symptom longevity in both premature and full-term newborns. Such derivation will also require validation in additional cohorts and would likely be enhanced by inclusion of other factors known to influence AD such as gut dysbiosis, host intrinsic factors, and/or skin dysbiosis developing after the first week of life. If reproducible in validation and more diverse cohorts, these findings offer promise for using microbiome data to predict high risk newborns that may benefit from preventative interventions. However, while these data do not yet provide for a predictive model, they reproduce and expand upon the evidence that dysbiotic signals pre-date, rather than merely reflect, the onset of AD-associated inflammation and may be enriched in metabolism of *Proteobacteria* species and sphingolipid metabolism.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

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Ethics approval and consent to participate

All subjects provided signed informed consent prior to collection and storage of samples. These studies were approved by the IRB of Inova Fairfax Hospital and The National Institutes of Health. Consent forms included language authorizing publication of findings. The authors have no competing interests. This work was supported by the Intramural Research Program of the National Institute of Allergy and Infectious Diseases of the National Institutes of Health (Myles, Hourigan).

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What's already known about this topic?

- Atopic dermatitis (AD) is associated with dysbiosis of the skin bacteria.
- One small study using only 16S bacterial taxonomy suggested that the dysbiosis associated with AD may predate symptom onset.
- A larger study following 281 infants for 1 year found global microbiome differences were associated with lower transepidermal water loss, but did not identify any taxa which might predict AD.

What does this study add?

- Using a larger cohort, this study reproduces the prior findings indicating that the dysbiosis associated with atopic dermatitis predates diagnosis.
- Expansion to use of shotgun metagenomics identified *Prevotella*, *Candida*, and lipid metabolism of *Proteobacteria* as additional biomarkers of subsequent onset of AD.
- This work may foster primary prevention by using the microbiome signature from the first week of life to identify high risk newborns.

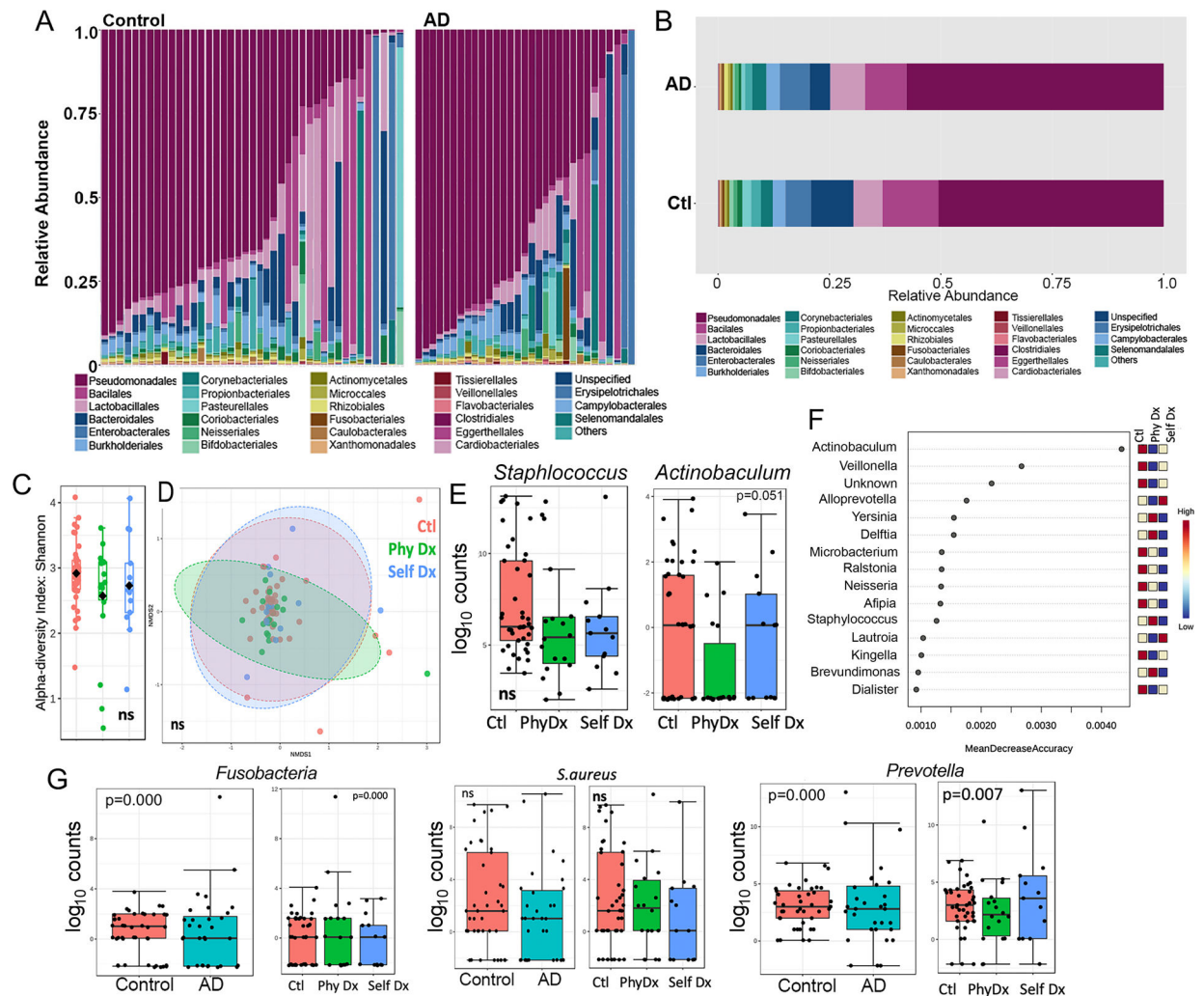


Figure 1: Bacterial taxonomic assessment for predicting atopic dermatitis.

(A-B) Order-level relative abundance of bacterial taxonomic assessment of ear skin swabs for individualized newborn participants (A) and group-wise (B) sorted by progression to atopic dermatitis (eczema) diagnosis or control. (C-D) Alpha diversity as measured by Shannon index (C) and NMDS visualization of beta diversity (using bray-curtis dissimilarity matrix) (D) for subjects further sorted by whether their diagnosis of AD was made by a physician (Phy Dx) or was self-diagnosed (Self Dx). (E) Genus level, log transformed count analysis for *Staphylococcus* and *Actinobaculum* relative abundance for participants as well as Random Forest (F) analysis for genus abundances. (G) Transformed count for genus or species level analysis for indicated taxa.

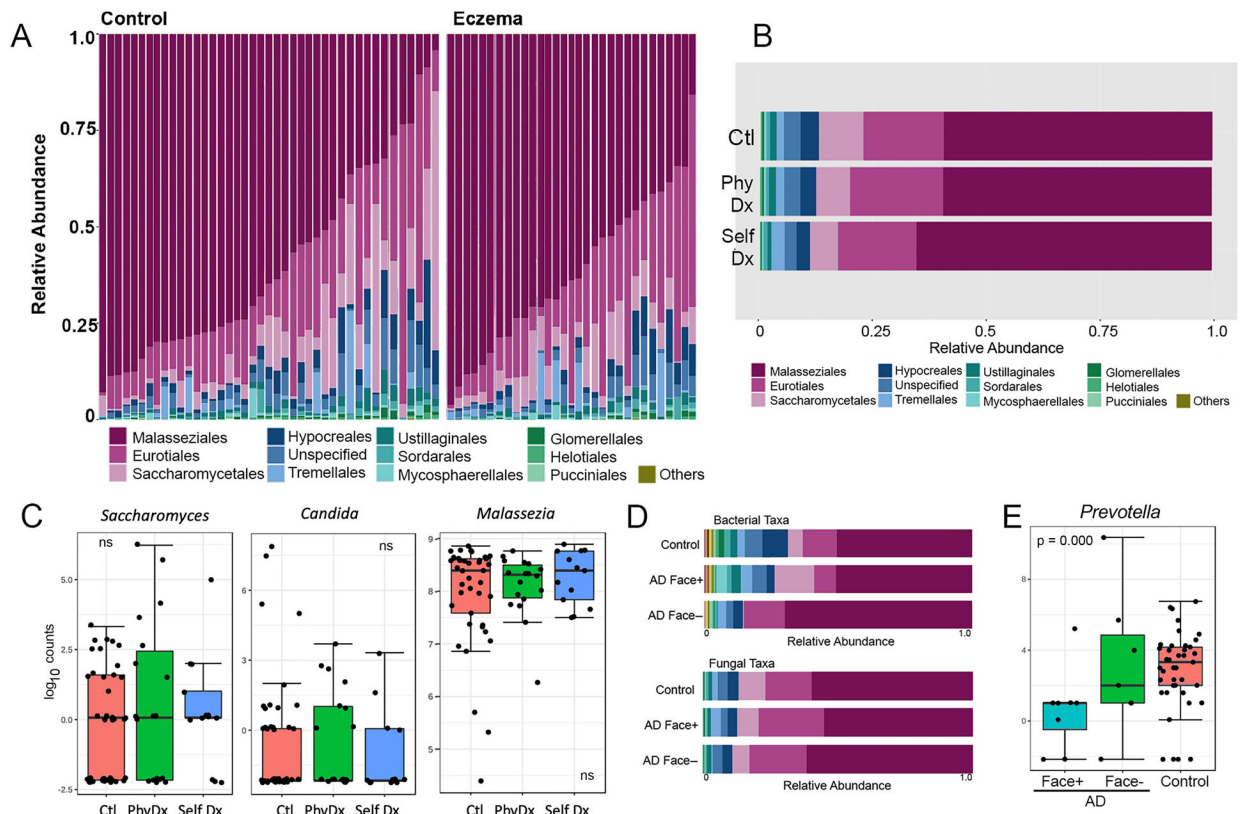
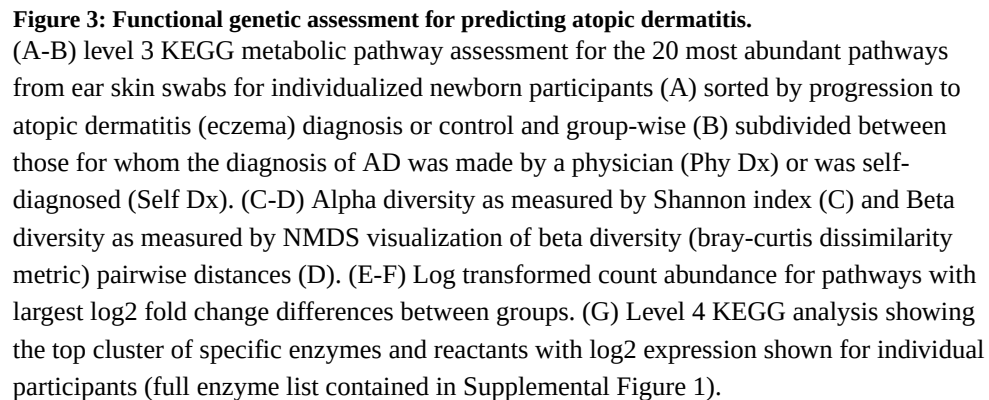


Figure 2: Expanded taxonomic assessment for predicting atopic dermatitis.

(A-B) Order level fungal taxonomic assessment for individual (A) and group wise (B) analysis. (C) Genus level comparison for indicated taxa. (D-E) Order level (D) and genus level (E) post-hoc analysis evaluating the 7 AD subjects reporting facial involvement (Face+), the 8 which reported AD but not on the face (Face-), versus controls; color coding for D is identical to the bacterial taxa in figures 1B and 2B.



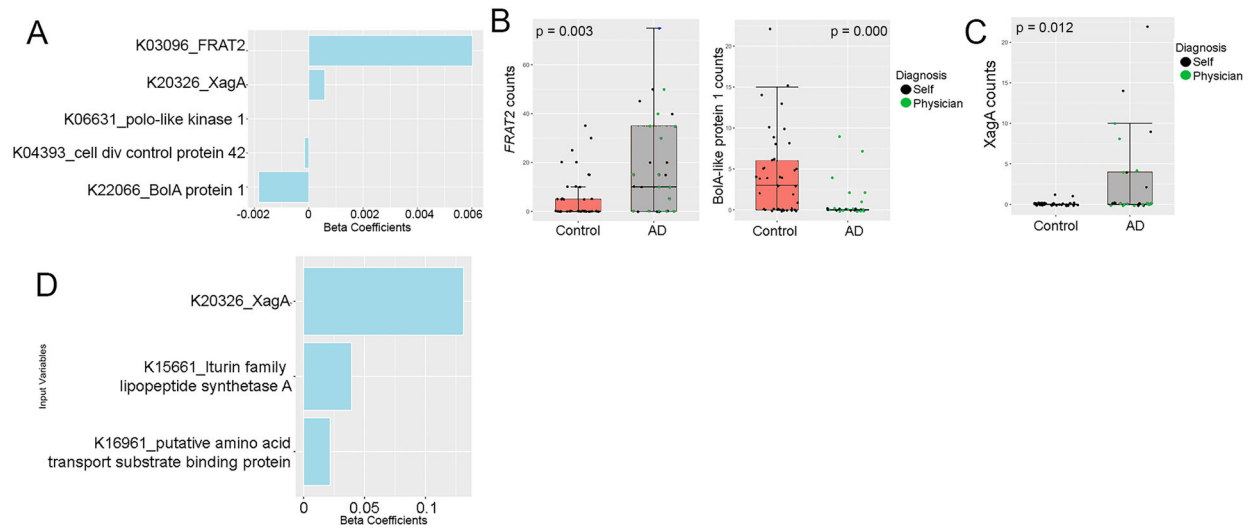


Figure 4: Regression for combined analysis predicting atopic dermatitis.

(A) Elastic net logistic regression analysis for combined bacterial taxa, fungal taxa, and level 4 KEGG analysis for distinguishing the control cohort from those that when on to develop AD by either self or physician diagnosis. (B) Individualized counts for indicated genes. (C-D) As in A-B but for factors distinguishing physician diagnosed AD versus controls.

Table 1:

Demographic and Clinical Features between those who developed AD and those who did not develop AD (controls).

	Atopic Dermatitis (n=31)	Control (n=41)	P value
Gestational Age (weeks)			
Mean	32.5	32.1	0.676
Range	24 - 40	23 - 39	
Delivery Mode (n, %)			
Vaginal	7, 23%	9, 22%	0.950
Cesarean	24, 77%	32, 78%	
Maternal antibiotics in labor (n, %)			
Yes	30, 97%	38, 93%	0.453
Infant antibiotics prior to skin swab (n, %)			
Yes	0, 0%	0, 0%	1
Maternal family history of skin conditions (n, %)			
Yes	2, 6%	7, 17%	0.178
Day of life of swab			
Mean	2.774	2.683	0.781
Range	1 - 5	0 - 7	
Ethnicity (n, %)			
Non-Hispanic or Latino	23, 74%	28, 68%	0.933
Hispanic or Latino	3, 10%	5, 12%	
Unknown	5, 16%	8, 20%	
Race (n, %)			
White or Caucasian	17, 55%	27, 66%	0.591
Black or African American	4, 13%	3, 7%	
Asian	4, 13%	8, 20%	
More than one Race	1, 3%	0, 0%	
Other	2, 6%	1, 2%	
Unknown	3, 10%	2, 5%	
Sex (n, %)			
Male	21, 68%	24, 59%	0.424
Female	10, 32%	17, 41%	